

Performance Testing

Date	30 June 2026
Team ID	XXXXXX
Project Name	RISING WATERS
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the Rising Waters system performs under different load conditions. It measures response time, resource utilization, stability, and reliability to ensure the application provides accurate flood predictions and real-time monitoring efficiently

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Locust, ML Testing
Type of Testing	Load Testing, Stress Testing, Performance Testing
Target Module / API	Flood Prediction, water Level Monitoring, Login
Test Environment	Local System (window)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	20	5 min	Login Successfully

2	Water Level Monitoring	30	5 min	Data loads currently
3	Flood Prediction	25	5 min	Prediction generates quickly
4				



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time	< 2 seconds	1.5 sec	Pass	Good
2	Average Load Time	< 5 seconds	3 sec	Pass	Acceptable
3	Throughput	High	High	Pass	Stable
4	Error Rate	< 1%	0.2%	Pass	Very Low
5	CPU Utilization	< 80%	65%	Pass	Normal
6	Memory Utilization	< 80%	60%	Pass	Stable

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

The Rising Waters application successfully handled user requests. Flood prediction results were generated accurately with minimal response time. The system remained stable during testing.

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Minor delay was observed when multiple prediction requests were submitted simultaneously. No major performance issues were identified.

Optimization Steps Taken:

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[Describe any changes or optimizations made to improve performance]

Optimized the Flask application, improved input validation, and reduced unnecessary processing to enhance performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

